CS4265 Milestone 2 Progress Report

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***Abstract*—This progress report summarizes the initial implementation of the proposed data pipeline developed for Milestone 2. It presents the components that have been successfully implemented, the challenges encountered during development, and the solutions applied or planned. The report also outlines modifications made since Milestone 1, describes the planned tasks for Milestone 3, and evaluates the current project timeline. In addition, the project directory structure and key evidence screenshots are provided to demonstrate the functionality and progress of the system.**

I. What Works

The following components of the data pipeline have been successfully implemented and tested:

- **fetch_data.py** – Successfully downloads the Online Retail dataset in Excel format from the official source and stores it in the local `data/raw` directory.
- **settings.yaml and .env.example** – Provide configuration management for the pipeline, including dataset paths, simulation parameters, and environment variable templates.
- **main.py** – Serves as the main orchestration script that coordinates the entire data pipeline, including data ingestion, processing, and storage stages.
- **db_handler.py** – Implements database interaction and stores the processed dataset into a SQLite database for structured storage and later analysis.
- **placeholder.py** – Acts as the initial scaffold for the data processing module, providing a structure where future data cleaning, transformation, and validation steps will be implemented.
- **Execution Evidence** – Screenshots demonstrating successful execution of the pipeline have been generated, including `fetch_data_success.png` and `main_output.png`, and stored in the `docs/` directory.
- **M2 Data Flow Diagram** – A visual representation of the pipeline architecture has been created and stored in the `docs/` directory, illustrating the flow from data ingestion to database storage.
- **requirements.txt** – Lists all required Python libraries to ensure the project environment can be reproduced.
- **README.md** – Provides project documentation, including setup instructions, pipeline description, and project structure.

II. Challenges Encountered

During the implementation of the data pipeline, several challenges were encountered:

- **YAML Formatting Issues** – YAML configuration files require strict formatting rules. Initially, tabs were used for indentation, which caused parsing errors. This was resolved by replacing tabs with spaces to comply with YAML syntax requirements.
- **Mermaid Diagram Formatting** – While creating the data flow diagram using Mermaid, several formatting adjustments were required to ensure compatibility with the Mermaid Live Editor and correct rendering of the pipeline structure.
- **Configuration and File Path Consistency** – Ensuring that the file paths defined in the configuration file (`settings.yaml`) matched the actual project directory structure required careful adjustments, especially after reorganizing the raw and processed data folders.

III. Changes from M1

Several improvements and extensions were implemented in Milestone 2 compared to the initial project structure developed in Milestone 1:

- **Database Storage Implementation** – A new module, `db_handler.py`, was added to support storing processed data into a SQLite database for structured storage and later analytical tasks.
- **Pipeline Orchestration** – The main pipeline execution logic was implemented in `main.py`, which coordinates the different pipeline stages including data ingestion, data expansion for big data simulation, basic data processing, and database storage.
- **Improved Project Structure** – The project directory structure was refined to better represent the stages of the data pipeline, separating raw data, processed data, configuration files, documentation, and source code modules.

IV. Next Steps for M3

- Implement data cleaning and validation in the processing module
- Extend pipeline for analytics and visualization
- Add automated testing for pipeline modules

V. Timeline Assessment

The project is on track. Initial implementation completed; pipeline tested end-to-end. Ready for M3 development.

VI. Project Structure

```
CS4265_Project_Jia_Liu/
+-- config/
|   +-- settings.yaml
|   +-- env.example
+-- src/
|   +-- ingestion/
|   |   +-- fetch_data.py
|   +-- processing/
|   |   +-- placeholder.py
|   +-- storage/
|       +-- db_handler.py
|   +-- main.py
+-- data/
|   +-- raw/
|   |   +-- online_retail.xlsx
|   +-- processed/
|   |   +-- online_retail_big.csv
|   |   +-- retail.db
+-- docs/
|   +-- M2DataFlowDiagram.png
|   +-- fetch_data_success.png
|   +-- big_data_simulation.png
|   +-- data_processing.png
|   +-- db_saved.png
|   +-- pipeline_output.png
+-- requirements.txt
+-- README.md
```

VII. Evidence Screenshots

```
(.venv) cindyliu@Cindys-MacBook-Air CS4265_Pr
Downloading dataset...
Data downloaded successfully to: data/raw/onl
```

Fig. 1. Successful execution of `fetch_data.py`: the Online Retail dataset was downloaded and saved in the `data/raw` folder.

```
Expanding dataset to simulate Big Data...
Expanded dataset saved to data/processed/online_retail_big.csv, shape: (5419090,
 8)
```

Fig. 2. Big data simulation step: the raw dataset is expanded to generate a larger CSV for testing pipeline scalability.

```
Processing data...
Data shape: (5419090, 8)
First 20 rows:
   InvoiceNo StockCode  ... CustomerID         Country
0     536365    85123A  ...    17850.0  United Kingdom
1     536365     71053  ...    17850.0  United Kingdom
2     536365    84406B  ...    17850.0  United Kingdom
3     536365    84029G  ...    17850.0  United Kingdom
4     536365    84029E  ...    17850.0  United Kingdom
5     536365     22752  ...    17850.0  United Kingdom
6     536365     21730  ...    17850.0  United Kingdom
7     536366     22633  ...    17850.0  United Kingdom
8     536366     22632  ...    17850.0  United Kingdom
9     536367     84879  ...    13047.0  United Kingdom
10    536367     22745  ...    13047.0  United Kingdom
11    536367     22748  ...    13047.0  United Kingdom
12    536367     22749  ...    13047.0  United Kingdom
13    536367     22310  ...    13047.0  United Kingdom
14    536367     84969  ...    13047.0  United Kingdom
15    536367     22623  ...    13047.0  United Kingdom
16    536367     22622  ...    13047.0  United Kingdom
17    536367     21754  ...    13047.0  United Kingdom
18    536367     21755  ...    13047.0  United Kingdom
19    536367     21777  ...    13047.0  United Kingdom

[20 rows x 8 columns]
```

Fig. 3. Basic data processing preview: displays the shape and first 20 rows of the expanded dataset after initial processing.

```
Saving to database...
Data saved to database: data/processed/ret
```

Fig. 4. Database storage confirmation: processed data has been successfully saved into the SQLite database at `data/processed/retail.db`.

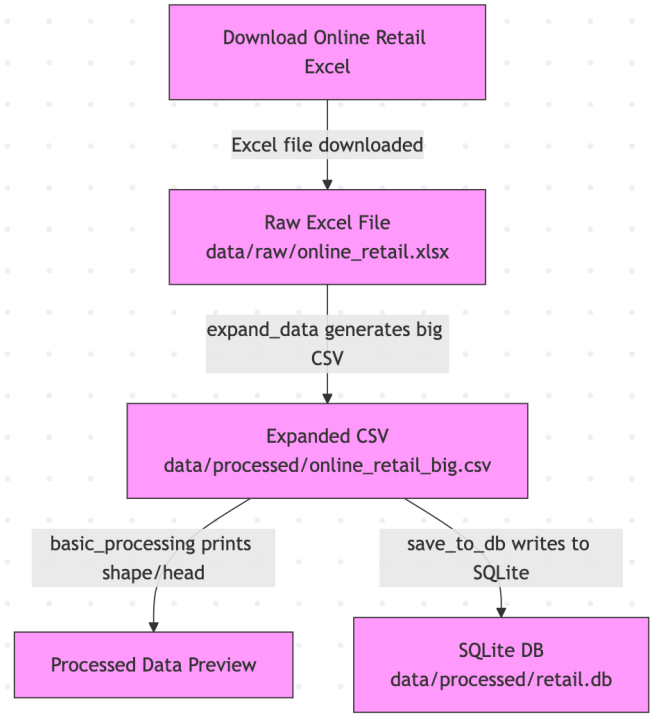

Fig. 5. M2 Data Flow Diagram: visual representation of the pipeline architecture from dataset download, data expansion, basic processing, to database storage.